

Explanation Loops I

What happens when you run the code? Explain here how a loop produces the result, and give yourself time to understand the logic. This code draws multiple circles on a black background at predefined positions. Each circle is filled with a random color, and the process repeats every frame (due to the `draw()` function), although the positions and colors remain constant unless the `xs` or `ys` arrays are changed dynamically during execution.

1. Global Variables:

- `xs` and `ys` are arrays (lists) that store the x and y coordinates for the circles. Each pair of values from these arrays will be used to draw a circle on the canvas.

2. `setup()` Function:

- The `setup()` function is called once when the program starts. It creates a canvas of 400x400 pixels and sets the background color to black.

3. `drawCircle(x, y)` Function:

- This is a custom function designed to draw a single circle at the specified (x, y) coordinates.
- Inside this function:
 - `noStroke()` is called to ensure the circles don't have an outline.
 - `random(256)` is used to generate random values for the red (`bubble_R`), green (`bubble_G`), and blue (`bubble_B`) color components, resulting in a random color.
 - `fill(bubble_R, bubble_G, bubble_B)` sets the fill color for the circle using the randomly generated RGB values.
 - `circle(x, y, 25)` draws a circle at the specified coordinates with a diameter of 25 pixels.

4. `draw()` Function:

- The `draw()` function is called repeatedly, continuously executing the code within it.
- Inside `draw()`, a `for` loop iterates over the `xs` array. For each iteration, it retrieves the corresponding x and y coordinates from `xs` and `ys` arrays and passes them to the `drawCircle()` function to draw a circle.
- As a result, a set of circles is drawn on the canvas at the specified coordinates, each with a random color.

More with Lists and For Loops II

The following code snippet has a few changes: We now use a loop to generate our list! Paste the code into a new project and consider the differences in the various loops.

```
// Initialize Lists
// Empty array to hold x-coordinates for the circles
let circleList = [];

function setup() {
  createCanvas(400, 400);
  background(0);
}

function drawCircle(x, y) {
  noStroke();
  // Generate a random colors value between 0 and 255
  bubble_R = random(255);
  bubble_G = random(255);
  bubble_B = random(255);
  // Set the fill color using the random RGB values
  fill(bubble_R, bubble_G, bubble_B);
  circle(x, 200, 50); // Draw a circle
}

function createCircleList(n) {
  // Add a new x-coordinate to the circleList array, spaced by 60 pixels
  for (var i = 0; i < n; i++) {
    append(circleList, i * 60);
  }
}

function draw() {
  // Draw a circle at each x-coordinate in the circleList, with y fixed at 200
  createCircleList(10); // Populate the circleList with 10 x-coordinates
  for (var i = 0; i < circleList.length; i++) {
    drawCircle(circleList[i], 200);
  }
}
```

Explanation for Loops II

- **createCircleList(n) Function:**
 - A loop runs n times, appending new values to circleList.
 - Each value added is $i * 60$, where i is the loop counter, resulting in x-coordinates spaced 60 pixels apart.
- **draw() Function:**
 - createCircleList(10): Calls the createCircleList() function to generate 10 x-coordinates and store them in circleList.
 - A loop then iterates over circleList, calling drawCircle() for each x-coordinate, with the y-coordinate fixed at 200 pixels.
 - This results in 10 circles being drawn on the canvas.